

A later-literature answer to the $k = 2$ subcase of Conjecture 5.2 in arXiv:1308.4718

Literature-status packet for run `fa_banach_001`

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Status. `literature_implied_answer_partial_subcase`. This note records a known later result, not a new solution produced by the run.

Source question

Balan–Wang consider, on PDF page 16 of arXiv:1308.4718v1, an $n \times (n + k)$ real matrix

$$F = [g_1, \dots, g_{n+k}], \quad \|g_j\| \leq 1,$$

and define

$$\tau(F) = \min_{\substack{S \subseteq \{1, \dots, n+k\} \\ |S|=n}} \sigma_n(F_S).$$

Their Conjecture 5.2 asks whether, for fixed k , there is a constant $C = C(k)$ such that

$$\tau(F) \leq \frac{C}{n^{k-1/2}}. \tag{1}$$

Later theorem and exact identification

Liu–Wang prove in Lemma 2.4 of their 2016 paper that every real $(n + 2) \times n$ matrix A whose rows have norm at most one satisfies

$$\min_{\substack{S \subseteq \{1, \dots, n+2\} \\ |S|=n}} \sigma_n(A_S) \leq \frac{C}{n^{3/2}} \tag{2}$$

for an absolute constant C .

Take $k = 2$ in the source and set $A = F^T$. The rows of A are precisely the columns g_j of F , so they have norm at most one. For every n -element index set S ,

$$A_S = F_S^T \quad \text{and hence} \quad \sigma_n(A_S) = \sigma_n(F_S).$$

Thus (2) is exactly (1) for $k = 2$.

The later proof uses a two-dimensional orthonormal basis of the kernel and reduces the estimate to finding two nearby projective directions on the circle. This is the same direct kernel-duality route suggested by the source problem. Liu’s 2024 paper cites both the source and the 2016 paper and gives a sharp version of the planar two-column extremal bound.

Scope and provenance

This is a complete answer only for the $k = 2$ subcase. It does not settle Conjecture 5.2 for arbitrary fixed k and says nothing about Conjecture 5.1. The 2016 paper’s higher-dimensional discussion reduces the next case to Heilbronn-type incidence geometry and gives partial results under additional geometric hypotheses.

The relation is classified as a literature-implied answer rather than an explicitly announced solution: the later paper proves the identical matrix inequality but does not refer to it by the source’s conjecture label. Yang Wang is a coauthor of both papers.

References

- [1] R. Balan and Y. Wang, *Invertibility and Robustness of Phaseless Reconstruction*, arXiv:1308.4718v1, Section 5, Conjecture 5.2, PDF page 16; later published in Appl. Comput. Harmon. Anal. 38 (2015), 469–488.
- [2] Y. Liu and Y. Wang, *On the Decay of the Smallest Singular Value of Submatrices of Rectangular Matrices*, Asian-European J. Math. 9 (2016), article 1650075, doi:10.1142/S1793557116500753, Lemma 2.4.
- [3] Y. Liu, *On Minimal Smallest Singular Value of Subframes for Signal Recovery*, Operators and Matrices 18 (2024), 273–293, doi:10.7153/oam-2024-18-17, Theorem 2.7.